Next, we used these activated Mo-DCs to prime autologous and allogeneic T cells, resulting in increased secretion of IFNγ, TNF-α, and IL-10. T cells primed in this way were co-cultured with fresh non-transduced SK-MEL-147, resulting in tumor cell killing. Currently we are expanding these studies to include three-dimensional cultures of melanoma cell lines and also the use of human melanoma samples in patient derived organoid (PDO) models. We expect to reveal the influence of oncolysis on tumor infiltrating cells, including DCs, macrophages, NK and T cells.

Conclusion: Collectively, our results indicate that p14ARF and IFNβ delivered by our adenoviral system induced oncolysis in human melanoma cells accompanied by adaptive immune response activation and regulation.

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ONE S.T.E.P. TECHNIQUETM IS EFFICIENT IN HARVESTING MESENCHYMAL STEM CELLS FROM HUMAN ADIPOSE TISSUE

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Background: Conventional (mechanical) laser-assisted liposuction (LAL) produce an intense photothermal effect that induces adipocyte destruction. Absorption of lasers varies in tissues by their different composition (fat, water, hemoglobin), and strongly depends on different laser wavelengths along with used dosimetric parameters. The 1210-nm wavelength lasers are described to have great affinity for human adipose tissue due to tissue's high optical absorption coefficient at 1210 nm. Safety of LAL is well documented but few studies described the effect of LAL on human mesenchymal stem cells from adipose tissue (ASCs) and much less using 1210-nm lasers. Here we compare ASCs harvested by LAL with the those harvested by One S.T.E.P. techniqueTM - Selective Tissue Engineering Photostimulation (diode laser 1210 nm), which does not cause photo-thermal lipolysis effect in contrast to conventional laser-assisted method.

Methods: This study was approved by the Ethic Committee of the Institution. Fat tissue was obtained from 4 patients by traditional liposuction and from 4 patients in whom One S.T.E.P. techniqueTM was used. Cells were grown in DMEM with 10% FBS and antibiotics (100 μg/mL streptomycin,100 UI/mL penicillin) at 37°C in 5% CO_2 atmosphere. ASCs were characterized measuring specific antigen markers by flow cytometry. In vitro differentiation capability into osteoblasts, adipocytes, and chondrocytes were performed using defined media and evaluated by Alizarin Red, Oil red O and Alcian blue staining, respectively.

Results: Cells isolated from adipose tissue by both methods adhered to plastic and had a fibroblastoid morphology. Cells expressed CD29, CD90 and CD105 antigens, and lacked expression of CD14, CD34, CD45, CD80, CD117 and HLA-DR. These findings are in accordance with the definition of mesenchymal stem cells as defined by the ISCT. Moreover, cells were able to differentiate in chondrogenic, osteogenic, and adipogenic lineages under appropriated conditions.

Conclusion: Conventional laser lipolysis destroys both adipocytes and MSC through intense photothermal effect. The One STEP techniqueTM does not rely on lipolysis, and releases both preserved adipocytes and MSCs. In fact, here the use of One S.T.E.P techniqueTM for isolation of ASCs was shown to be an effective method to harvest ASCs cells, preserving their mesenchymal stem cell characteristics. Therefore, this method is potentially suitable for obtaining ASCs for using in regenerative medicine.

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OSTEOGENIC POTENTIAL OF NANOSTRUCTURED CARBONATED HYDROXIAPATITE MICROSPHERES ASSOCIATED WITH MESENCHYMAL STEM CELL IN VITRO AND IN VIVO

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Background: Bone tissue is capable of self-healing in the event of injuries. However, situations such as the patient's age or size of the damage can compromise natural healing, bringing to the fore the need for interventions to treat bone injuries. Tissue engineering presents treatment alternatives involving the use of cells and associated biomaterials. In this context, studies using biomaterials claim that carbonated hydroxyapatite (CHA) is more resorbable than hydroxyapatite (HA) and has biological similarity to bone hydroxyapatite.

Methods: Parameters of indirect contact (cytotoxicity, proliferation, and migration) of the extract obtained from spheres of carbonated hydroxyapatite associated with murine bone marrow stem cells (BMSCs) in vitro were evaluated. Then the osteogenic potential of CHA extract in murine BMSCs in vitro was assessed by quantification of calcium matrix, alkaline phosphatase levels and gene expression Col1A1 and Bglap. Non-critical size bone defects were performed in rat femur. After 14 days of healing, bone blocks were obtained for analysis from treated animals and compared to non-treated animals by histological and histomorphometric evaluation.

Results: The CHA extract showed osteoinductive properties in vitro without the need to add osteogenic inducing medium. The ability to induce osteogenic differentiation in undifferentiated cells was confirmed by increased activity of alkaline phosphatase, formation of mineralized nodules and increased expression of the Col1A1 and Bglap genes after 7 and 14 days of differentiation. All groups presented bone neoformation at the injury site 14 days after the operation. Histomorphometric analysis showed no statistical difference between groups in the parameters of bone formation, biomaterial and connective tissue.

Conclusion: The CHA spheres have in vitro osteoinductive properties and, when loaded with murine BMSCs, have the potential for bone repair and regeneration in vivo, presenting biocompatibility and osteoconductive, making it a promising alternative for bone tissue engineering applications.

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OSTEOGENIC, GENOTOXIC AND ANTIMICROBIAL POTENTIAL OF HYDROXYAPATITE AND ZN-HYDROXYAPATITE NANOPARTICLES ADDED TO AH PLUS ELUATES

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Background: Endodontic sealers are continuously being improved to enhance performance after root canal treatment. Despite their clinical success, commercial sealers do not meet the all the requirements of an ideal endodontic sealer, namely the induction capacity of the periradicular regenerative process that requires the recruitment of mesenchymal stem cells (MSC)/osteoblastic precursor cells and their differentiation into osteoblasts for synthesis of the mineralized matrix. Hydroxyapatite nanoparticles (NHAp) present similarities in size, crystallography, and chemical composition to the bone and teeth enamel.

Methods: The biological profile of NHAp and zinc-containing NHAp (NHAp-Zn), tested alone or added to the eluates of the endodontic sealer AH Plus, was characterized taking advantage of the biocompatibility of NHAp, the osteogenic effect of NHAp and zinc, together with the excellent profile of AH Plus. For this purpose, different cells were used: human mesenchymal stem cells derived from Wharton's jelly